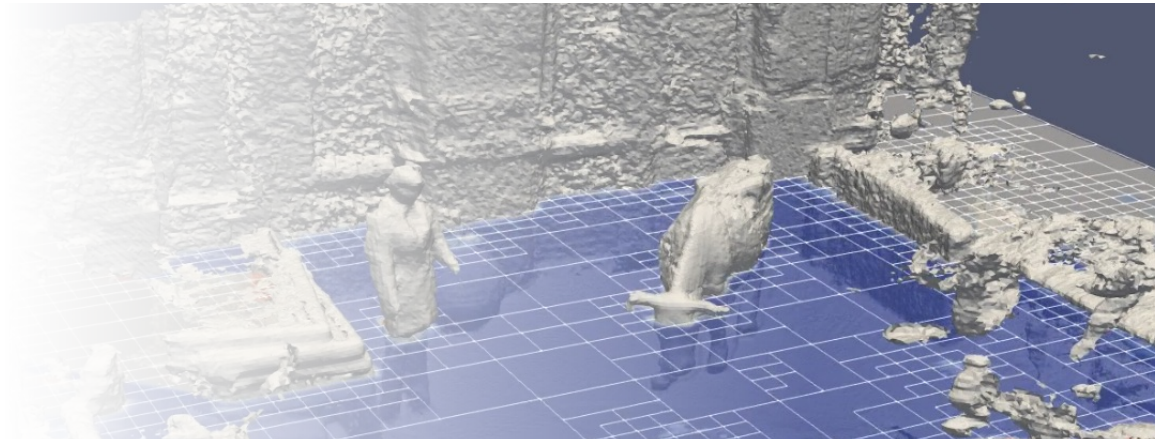


Mobile Robotics (IN2404)

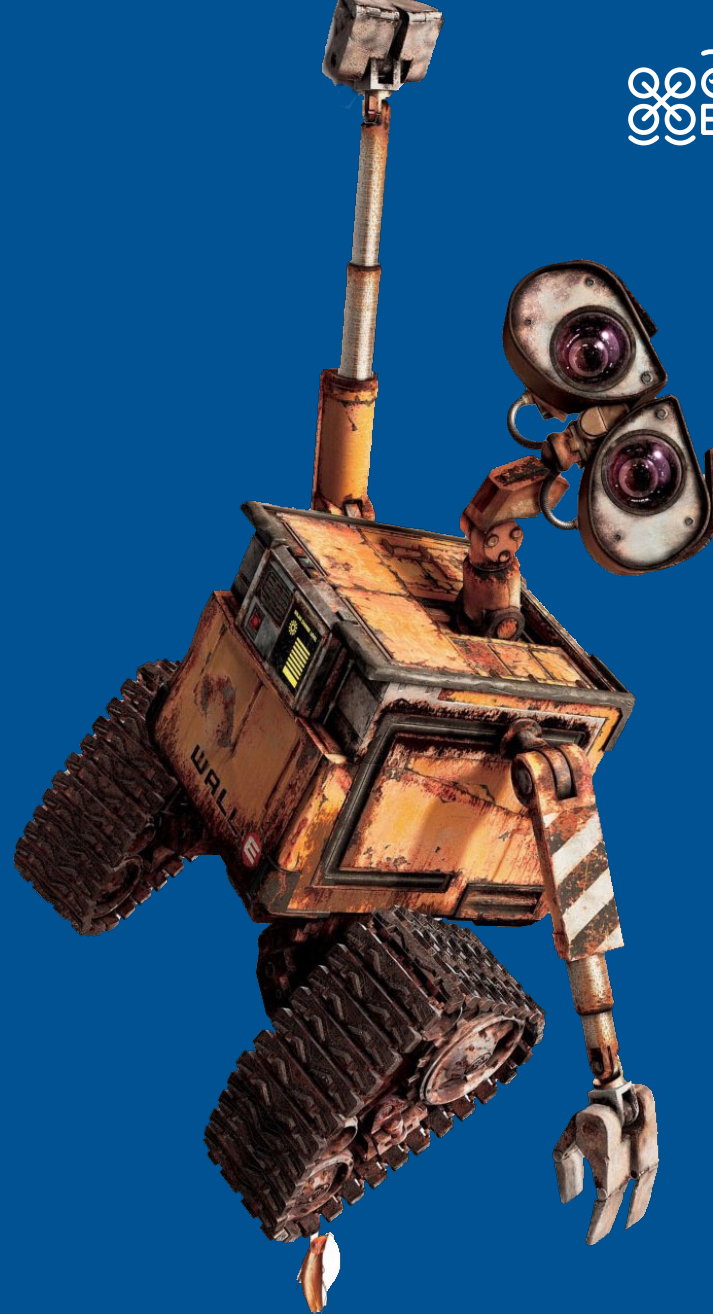
Lecture 1: Introduction

Prof. Dr. Stefan Leutenegger

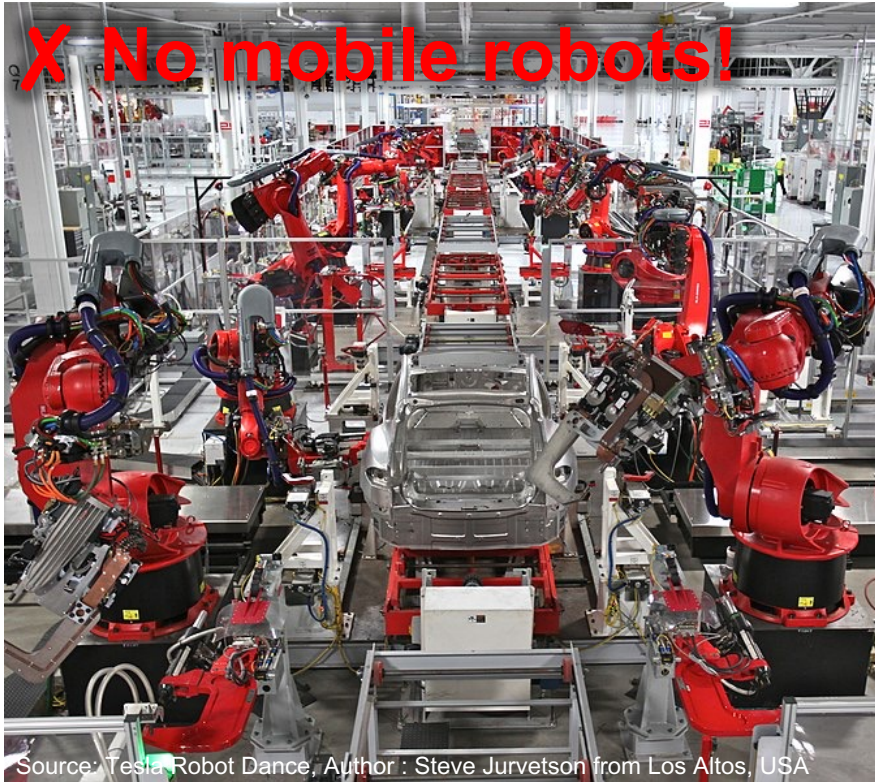
Teaching Assistants: Dr. Jaehyung Jung, Simon Schaefer, and Sebastián Barbas Laina



Introduction



What is a Mobile Robot?



Factory robots

- Stationary
- Controlled environment
- Clear, precise, repetitive tasks



Lawn mower robot

- Mobile: within lawn
- Semi-controlled env. (docking, fences)
- Clear task, open in implementation

(Non-Military) Mobile Robots Today



Domestic robots
(picture: Dyson 360 Eye)



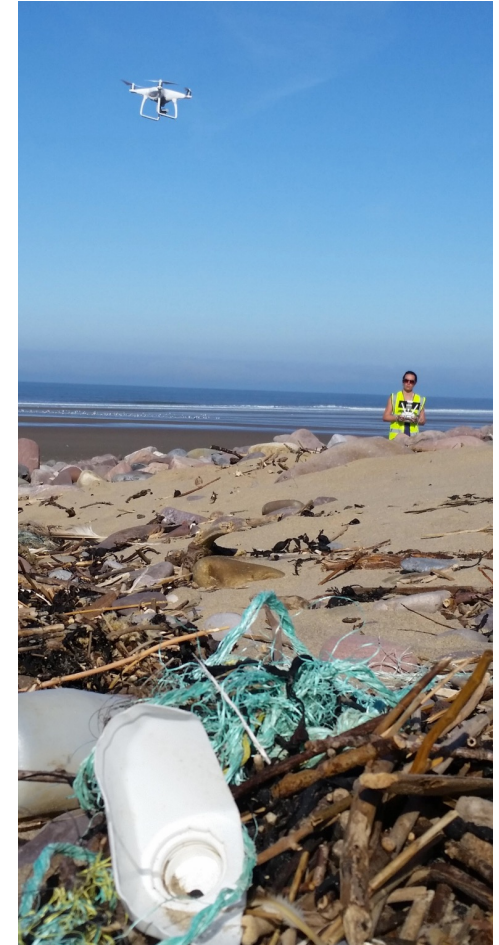
**Drone inspection
and construction**



Space exploration
(picture: NASA Perserverance and Ingenuity)



Self-driving cars
(picture: Waymo self-driving car)



**Environmental
monitoring**

Visual-Inertial SLAM & Model-Predictive Control

Barbas Laina, Boche, Papatheodorou, Tzoumanikas, Schaefer, Chen, Leutenegger

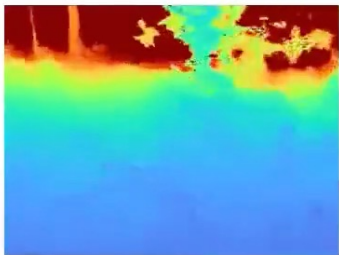
Real world experiment



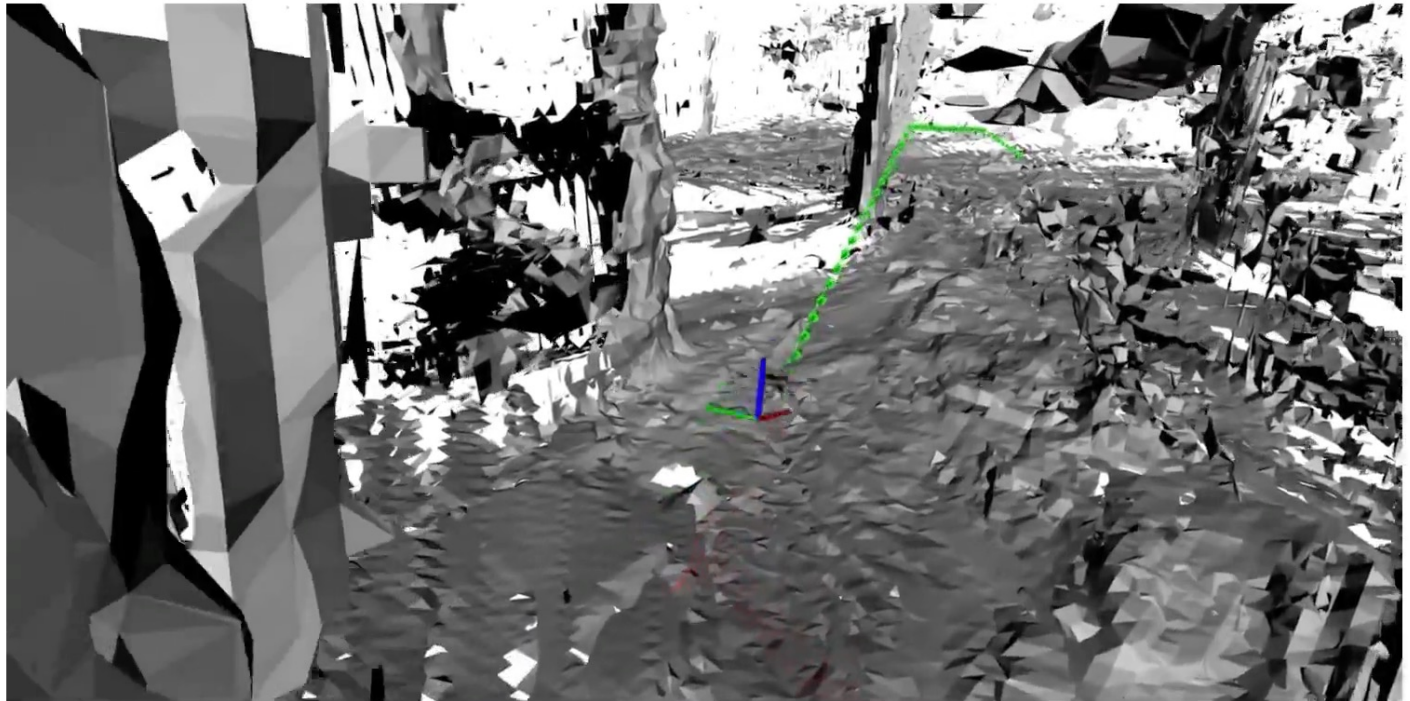
External view



MAV IR (15 Hz)



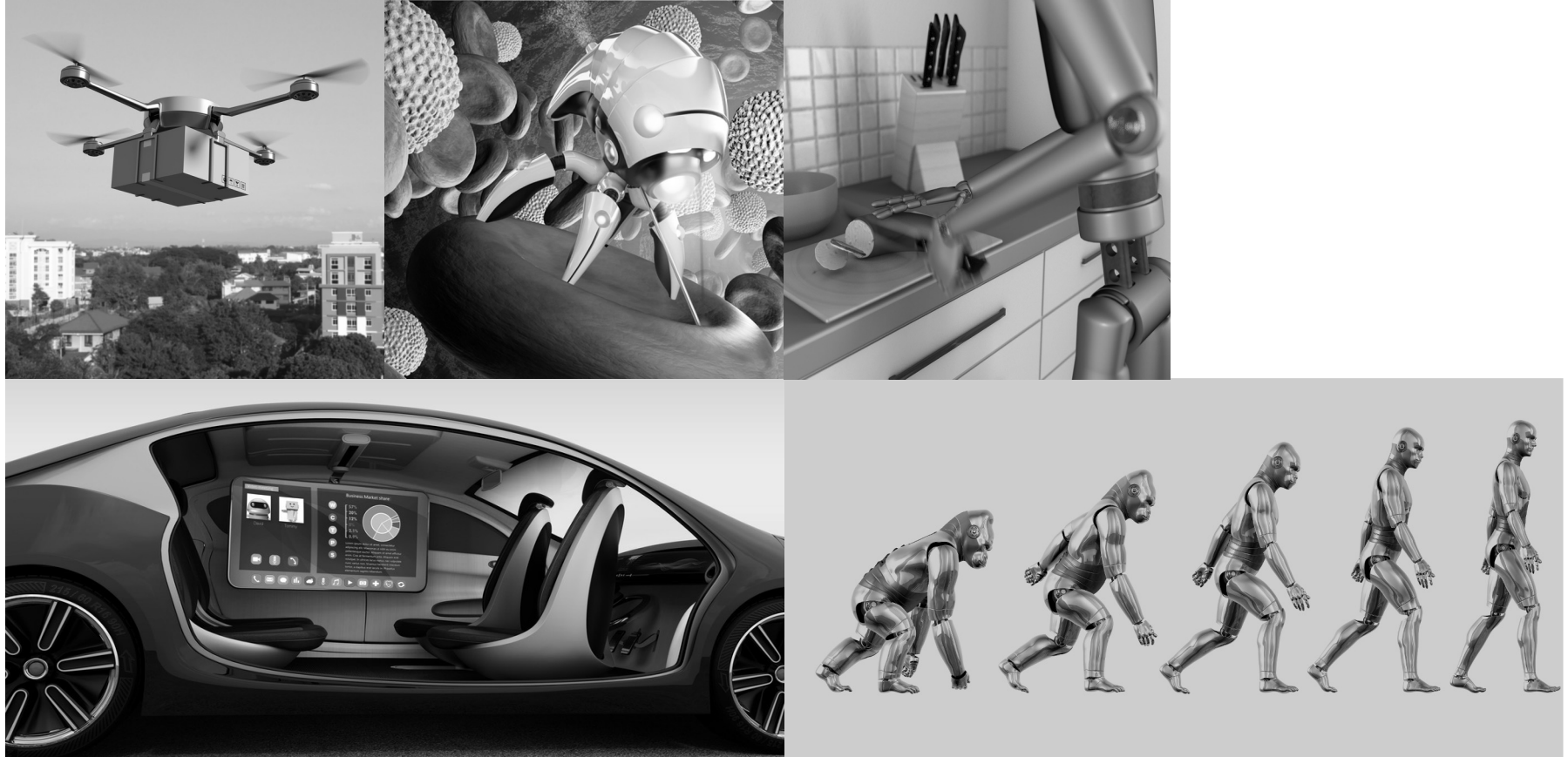
MAV depth (5 Hz)



MAV velocities up to 3 m/s

1x

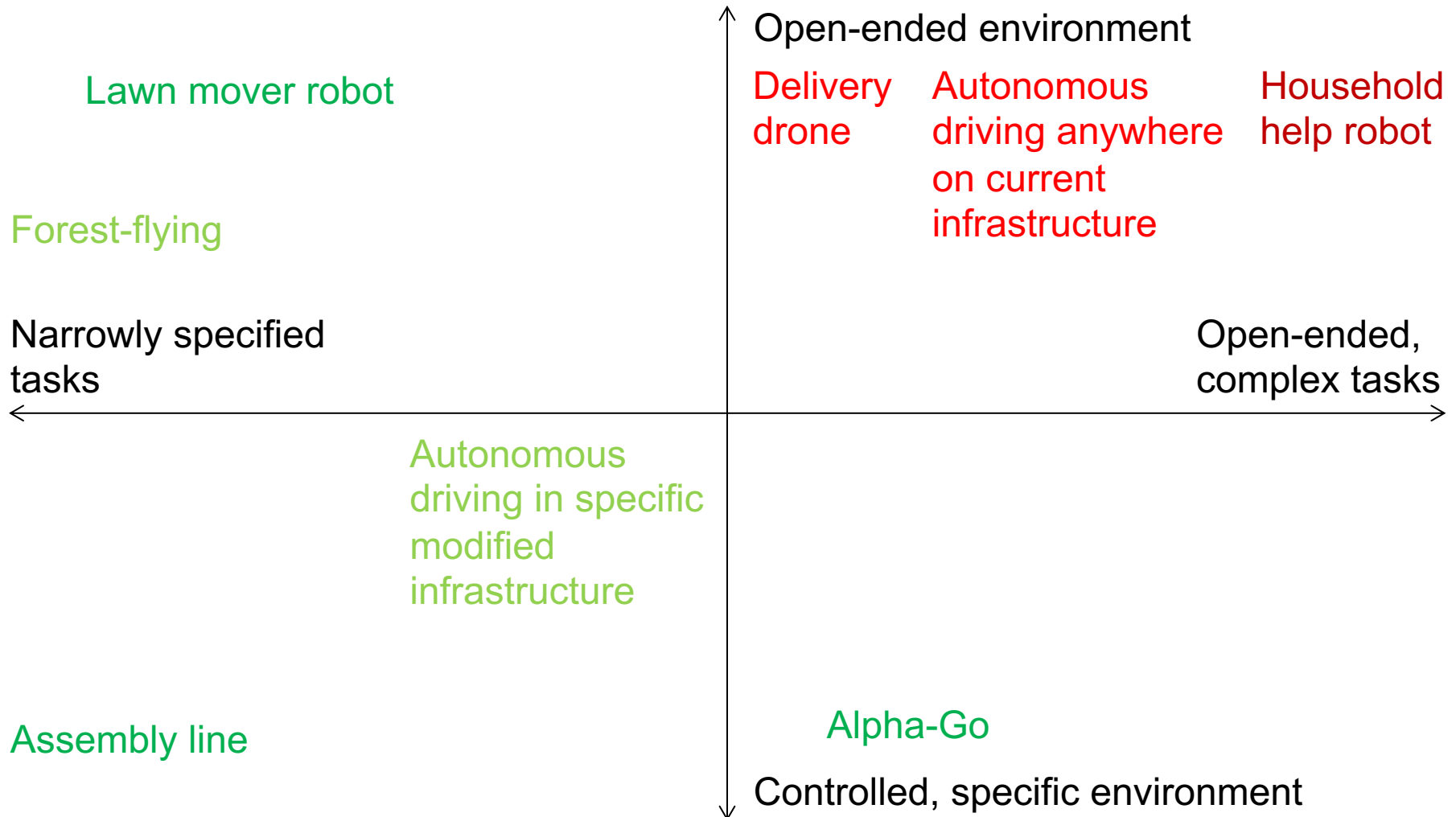
The Promised Future of Mobile Robots



Why would you want a mobile robot?

- Does boring/repetitive/time-consuming **chores** for us
- Can go into **dangerous** or even unlivable environments
- Carries out tasks in **remote** places that are impossible to reach

Readiness of (Mobile) Robots in 2023



Questions for the Mobile Robot



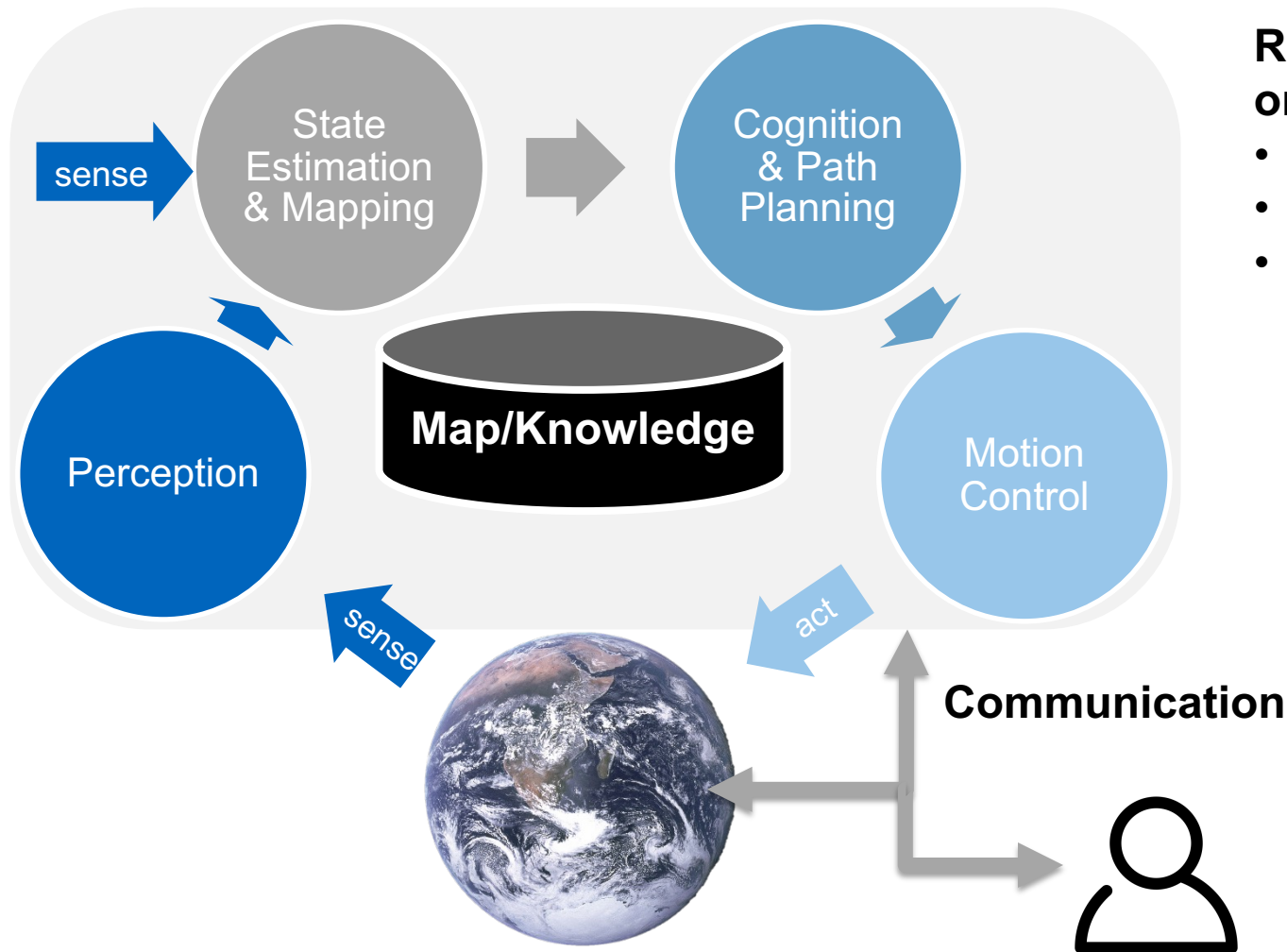
Where am I?

What does the world look like?

Where am I going?

How do I get there?

Mobile Robotics Modules and Challenges

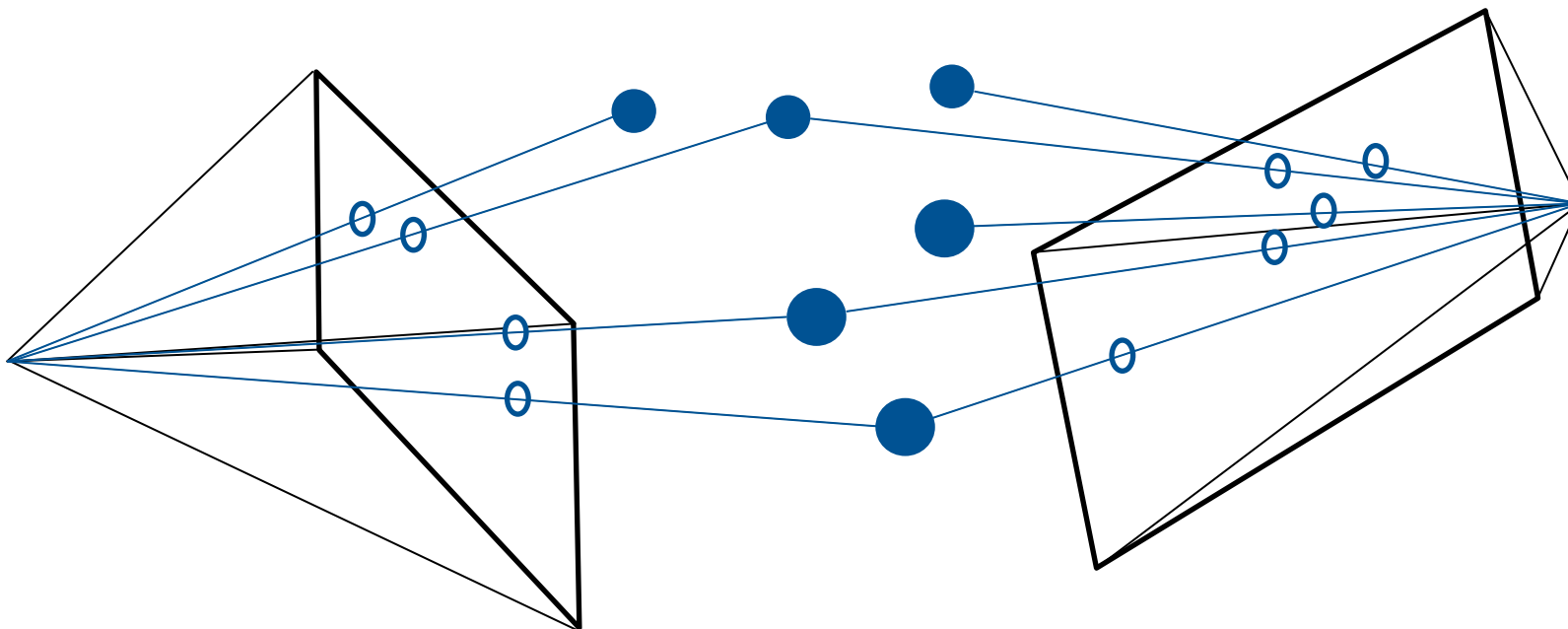


Requirements on algorithms:

- Robust
- Efficient
- Accurate

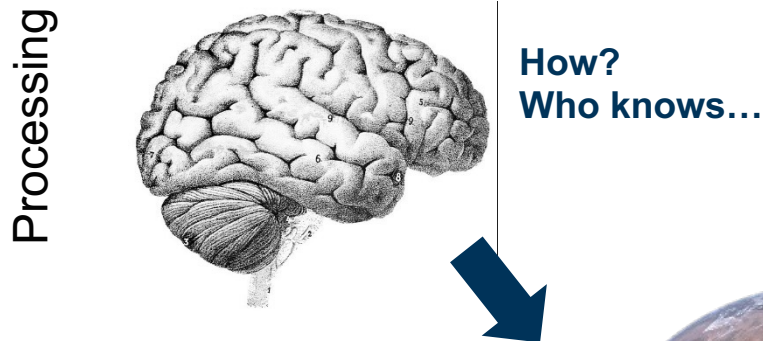
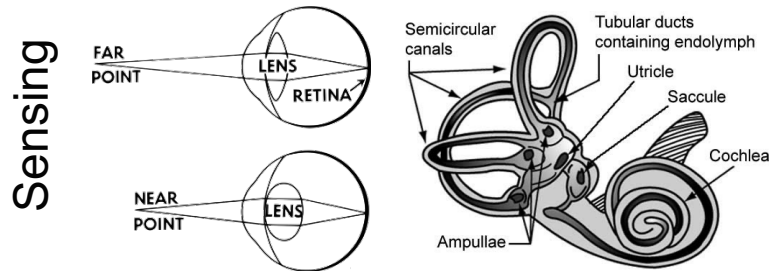
The Visual SLAM Problem

SLAM: Simultaneous Localisation and Mapping

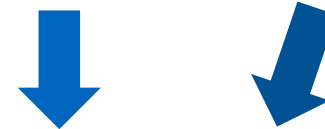
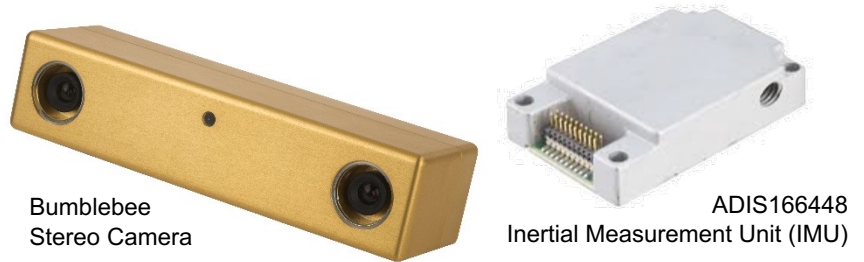


Why Multi-Sensor Fusion?

How humans do it



How robots do it



Using maths and code

$$\mathbf{b}_{\lambda_1}^* = \underbrace{\mathbf{b}_{\lambda_1,0} - \mathbf{H}_{\lambda_1\mu} \mathbf{H}_{\mu\mu}^{-1} \mathbf{b}_{\mu,0}}_{\mathbf{b}_{\lambda_1,0}^*} - \mathbf{H}_{\lambda_1\lambda_1}^* \Delta \chi_{\lambda_1}$$

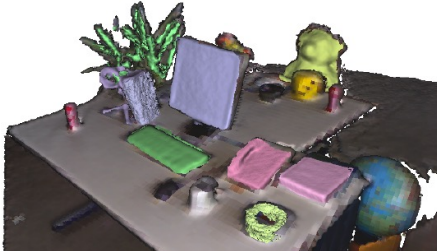
```
bool MarginalizationError::marginalizeOut(
    const std::vector<uint64_t> & parameterBlockIds,
    const std::vector<bool> & keepParameterBlocks)
{
    if (parameterBlockIds.size() == 0)
        return false;
}

// copy so we can manipulate
std::vector<uint64_t> parameterBlockIdsCopy = parameterBlockIds;
```

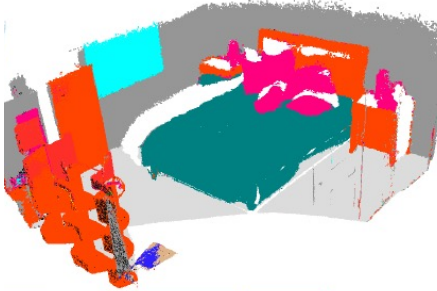


**Localisation
and Mapping**

Levels in Spatial AI



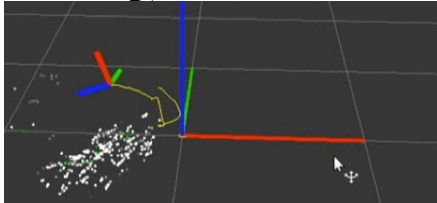
- **Object-level** and **dynamic maps** e.g. [MID-Fusion]
 - Advanced physical interaction robot-environment
 - More advanced human robot interaction



- **Semantic maps:** 3D scene understanding e.g. [SemanticFusion]
 - Advanced task planning
 - Advanced human robot interaction



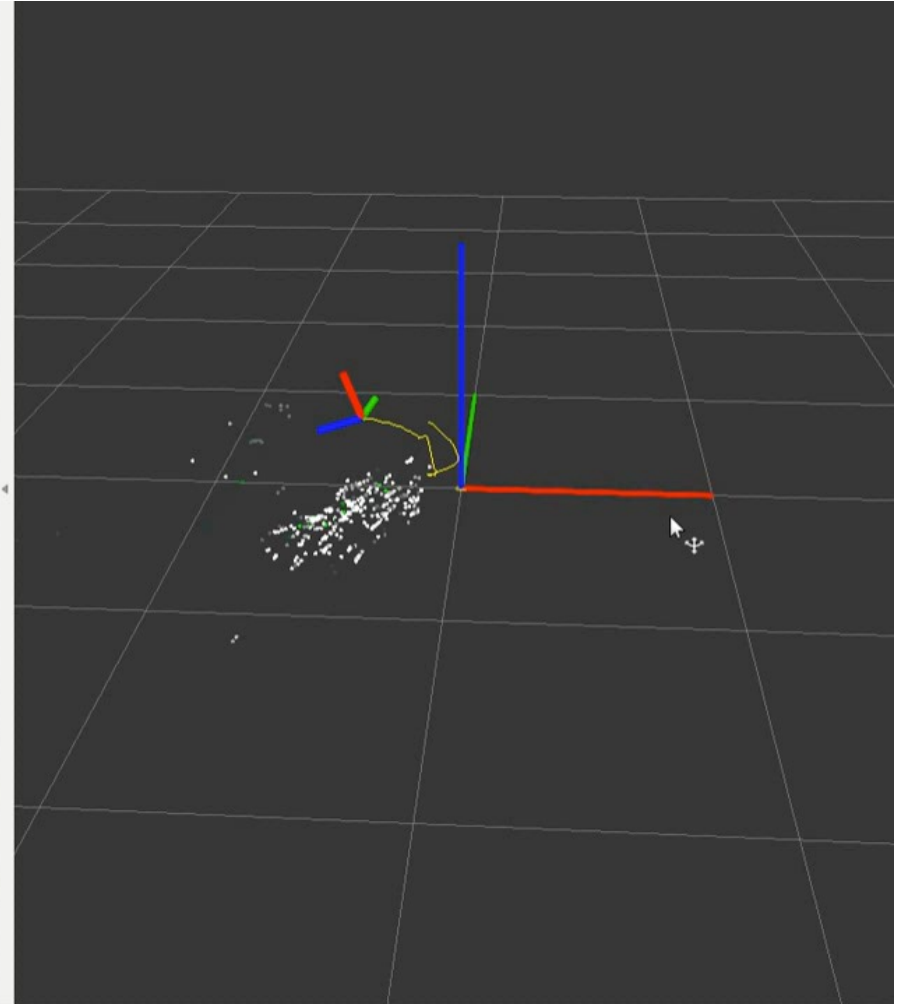
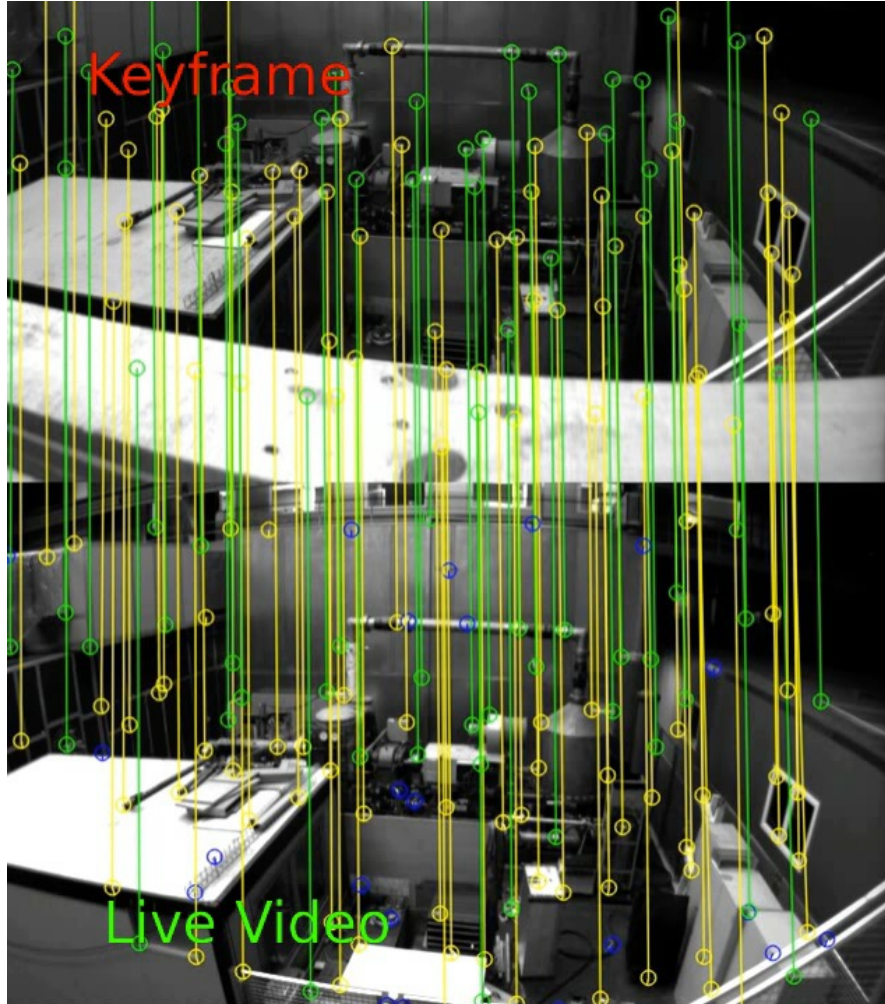
- **Dense mapping** e.g. [ElasticFusion, supereight]
 - Motion planning
 - Collision avoidance



- Sparse **Visual-Inertial** Localisation and Mapping e.g. [OKVIS, OKVIS 2]
 - Robust & accurate motion tracking for position control

OKVIS: Open Keyframe-based Visual-Inertial SLAM

Leutenegger, Furgale, Bosse, Lynen, Chli, Rabaud, Konolige, Siegwart



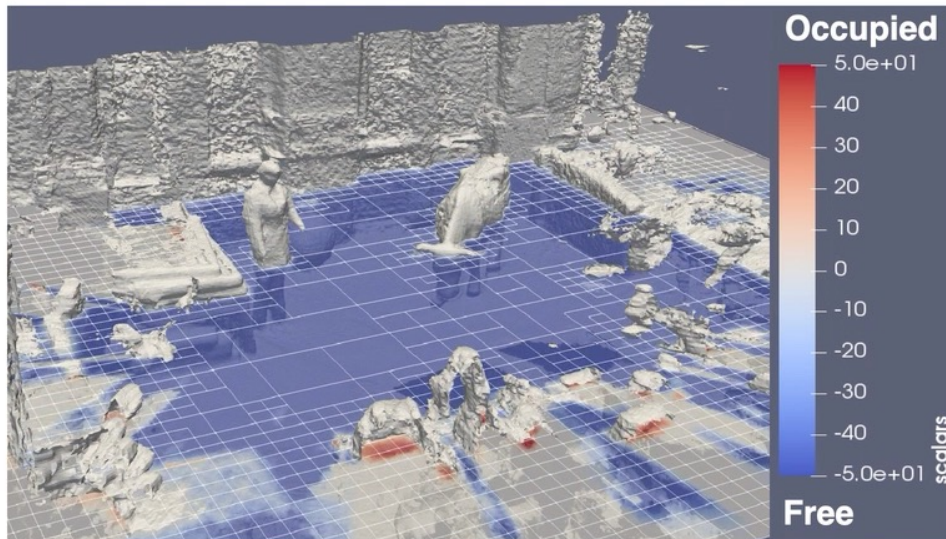
https://www.youtube.com/watch?v=TbKEPA2_-m4

From Dense Occupancy Mapping to Planning [ICRA'21]

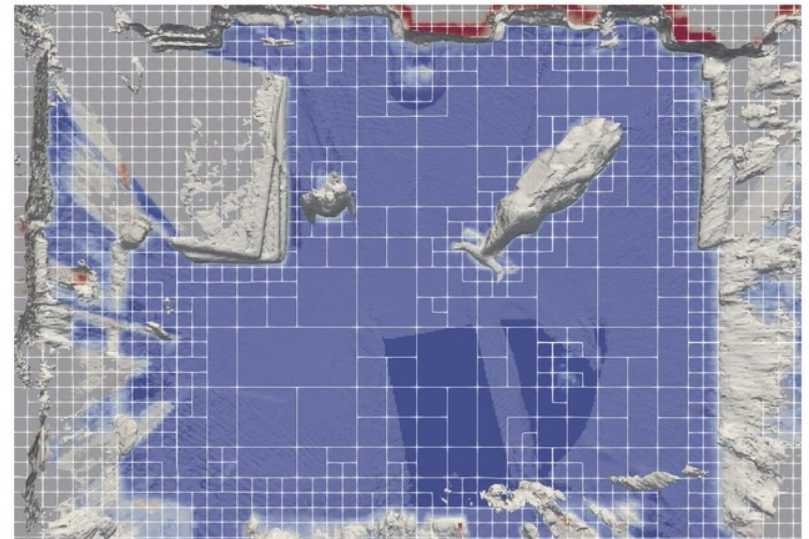
Funk, Tarrio, Papatheodorou, Popovic, Alcantarilla, Leutenegger

Map Visualisation

Side view



Top view

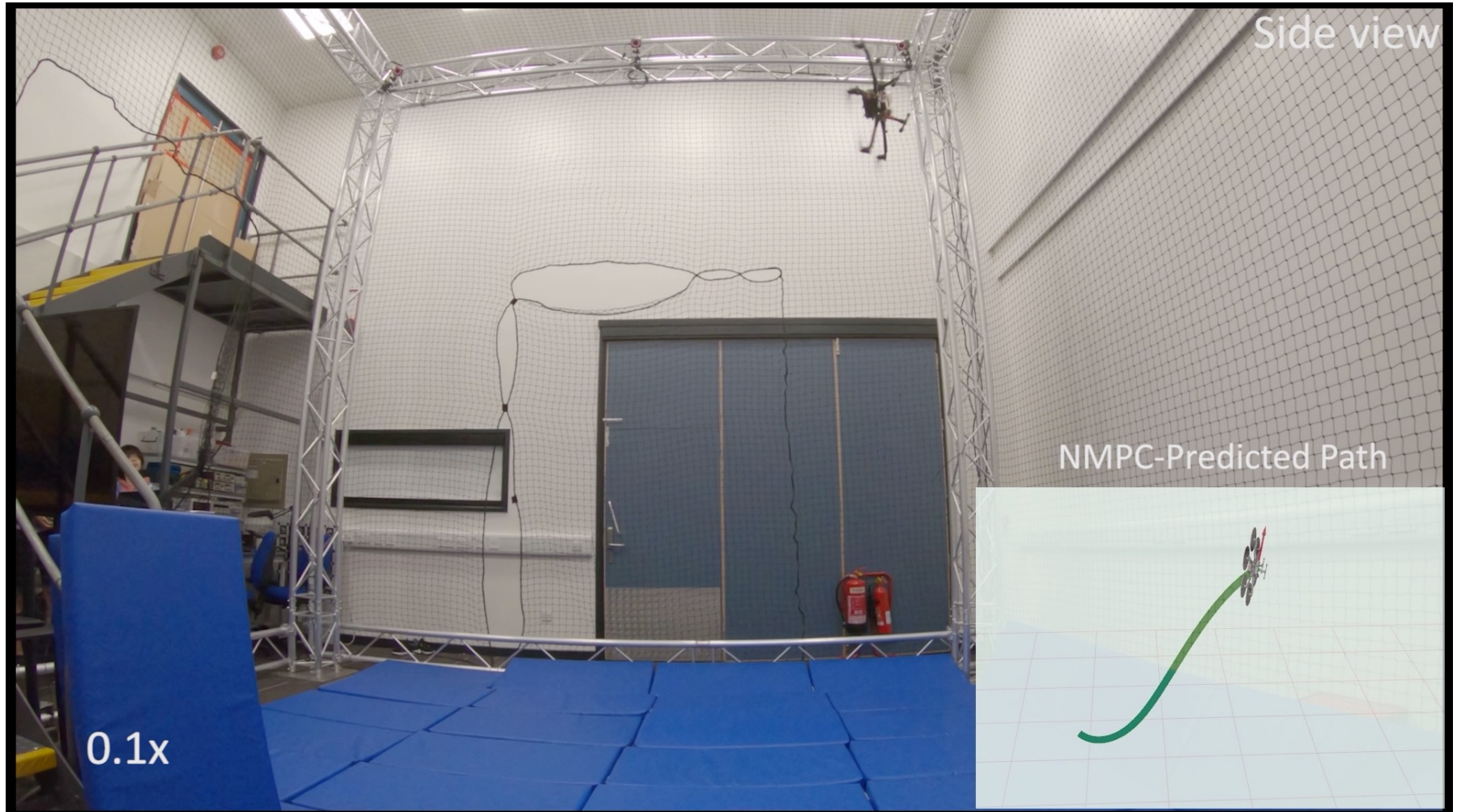


Explicit free-space representation encoding in a hierarchical way, supporting fast collision checking as crucial in robotic path planning and collision avoidance.

<https://youtu.be/XInO3GBNPvc>

Aggressive Flight / Failure Recovery [ICRA'20]

Tzoumanikas, Yan, Leutenegger



<https://youtu.be/cAQeSZ3tlqY>

Learning Outcomes

Upon successful completion of this module, students are able to

- describe the different **components** of mobile robotics, including their interactions, from **perception** to **estimation**, **mapping**, and **control**.
- work with different mathematical representations of **robot states** and **maps**.
- understand the different **probabilistic multi-sensor-fusion** approaches.
- understand **modern SLAM systems** including elements of **Machine Learning**.
- describe **navigation** algorithms including **motion planning** and **control**.
- create suitable mobile robotics algorithms for different **applications**.

Lecture Schedule

Date	Part	Contents
18/10/2023		Introduction
25/10/2023	I Foundations	Representations of States, Environment, and Uncertainty
01/11/2023		(No Lecture and no Exercise)
08/11/2023		Overview of Sensors and Actuators
15/11/2023	II Multi-Sensor Estimation	Probabilistic Nonlinear Least Squares
22/11/2023		Kinematics and Temporal Models
29/11/2023		Recursive Estimation
06/12/2023	III From SLAM to Spatial AI	Sparse Visual and Visual-Inertial SLAM
13/12/2023		Dense Mapping and SLAM
20/12/2023		Semantic, Object-level and Dynamic Mapping
10/01/2024	IV Planning and Control	Feedback Control
17/01/2024		Model-Based and Model-Predictive Control
24/01/2024		Motion Planning and Exploration
31/01/2024		Reinforcement Learning: Relation to Models, Ctrl & Perception
07/02/2024		Revision

Exercises

- Fridays 12:00-14:00 in 00.5901.051, Hörsaal (5901.EG.051)
(Electrical Engineering Building on the other side of the big parking lot)
- Tasks aligned with previous lecture
- Mix of tasks:
 - Descriptive, high-level understanding
 - Mathematical / algorithmic problem sets
 - Implementations in Python

The Team



Prof. Dr. Stefan Leutenegger

Phone: +49 89 289 17700

Office: 03.09.040

Email: stefan.Leutenegger@tum.de

Web: www.srl.in.tum.de

Teaching Assistants:



Dr Jaehyung
Jung



Simon
Schaefer



Sebastián
Barbas Laina

Course Material

Course Material: Moodle

- Schedule
- Slides
- Exercise task sheets

Literature (note: very extensive):

- [1] Probabilistic Robotics (2006),
by Sebastian Thrun, Wolfram Burgard, and Dieter Fox.
See also <http://www.probablistic-robotics.org/>
- [2] Estimation with Applications to Tracking and Navigation: Theory Algorithms and Software (2001),
by Yaakov Bar-Shalom, X. Rong Li, and Thiagalingam Kirubarajan.
- [3] Optimal State Estimation: Kalman, H Infinity, and Nonlinear Approaches (2006),
by Dan Simon.
- [4] Szeliski, Richard. Computer vision: algorithms and applications. Springer Science & Business Media, 2010.

Assessment

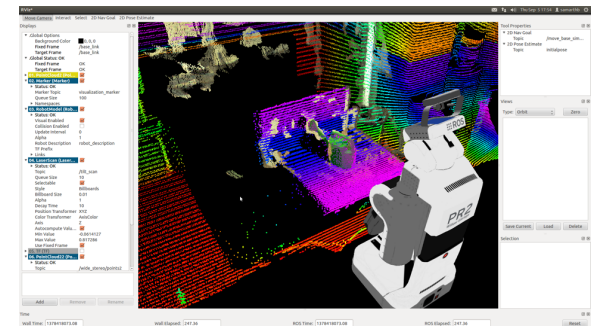
- 90 minutes written exam
- If only few students sign up, it may be done as oral exam instead.

Mobile Robotics Practicals

Running in parallel to this module.
Please sign up, if you are interested!

Goals

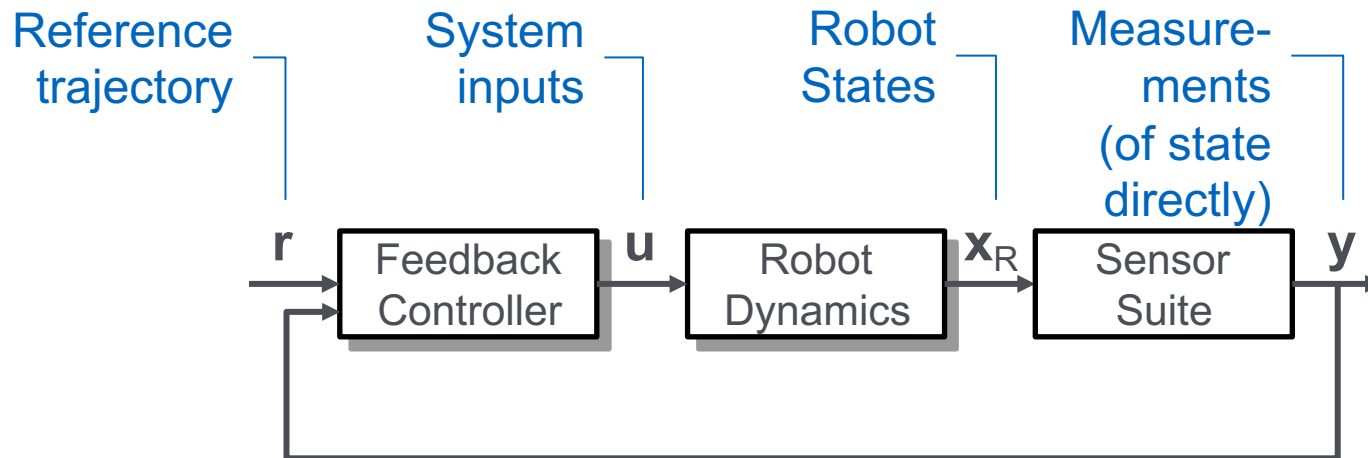
- Perform **navigation challenge** through cluttered environment with Parrot's AR Drone 2, see www.parrot.com/uk/drones/parrot-ardrone-20-elite-edition in **simulation** and the **real world**.
- Using the Robot Operating System (**ROS**) for communication and introspection, see www.ros.org
- Write generic and efficient **C++ code**, exploiting Inheritance, templates, and the Eigen library.
- Implement
 - Visual-inertial localisation
 - Dense volumetric mapping
 - Deep-learning-based obstacle recognition
 - Position feedback-control
 - Planning algorithms to navigate from start to goal



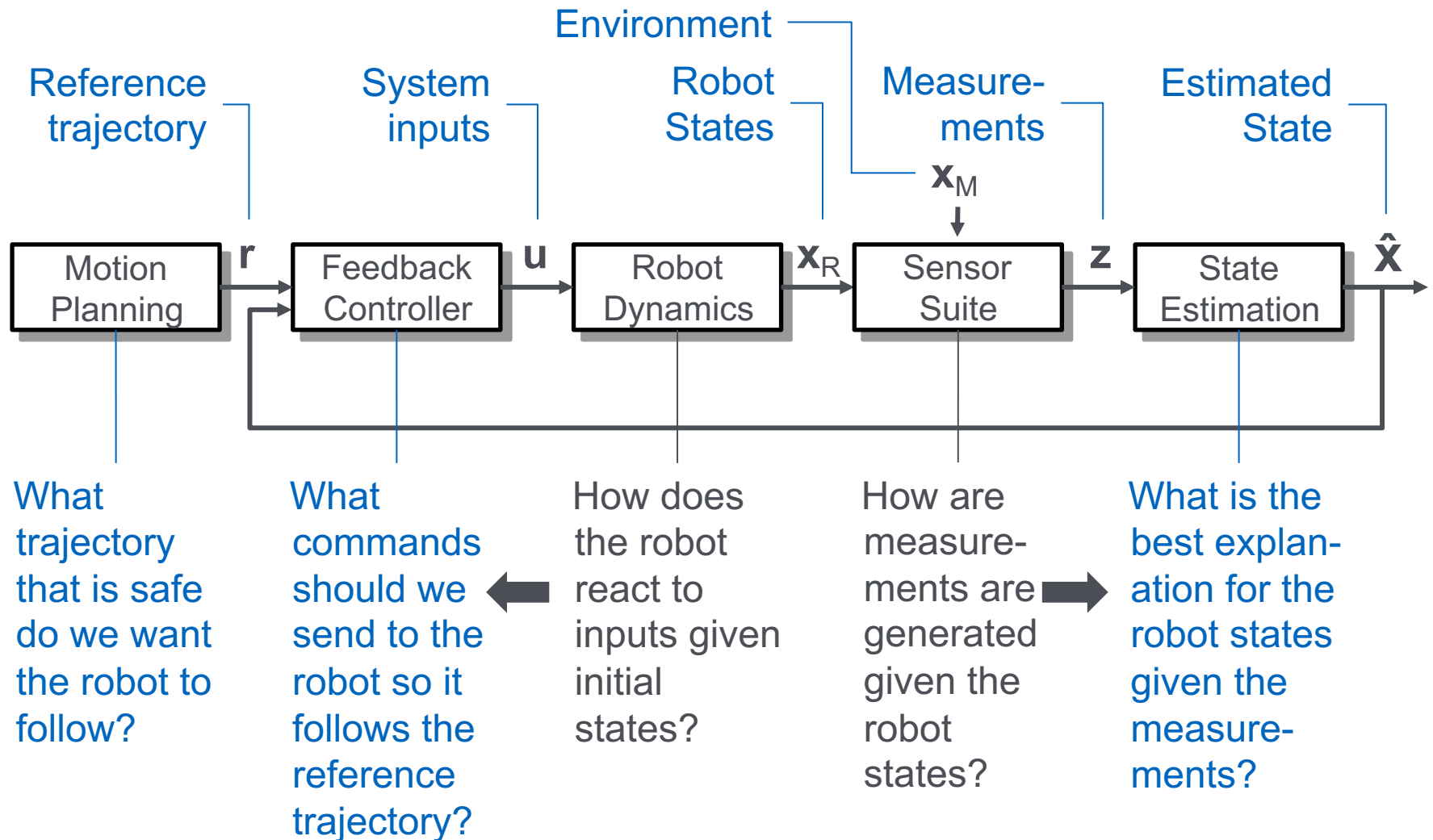
AR Drones: The Amazin' Challenge



A Relatively Simple Setting



A More Realistic Setting



To Be Estimated: Robot States and Environment

Robot state

Typical components:

- Position
- Orientation
- Velocity
- Sensor-internal states
- ...

Characteristics:

- Typically *time-varying*

Map – the environment

Popular representations:

3D points (sparse)

3D pointcloud/surfels (dense)

Occupancy grid / voxels

Triangulated mesh

...

Characteristics:

Typically, a large part is assumed to be *static*

Robot State

An example state vector:

$$\mathbf{x}_R^T = \left[\underbrace{W\mathbf{t}_S^T, \mathbf{q}_{WS}^T}_{\text{Physical robot states}}, \underbrace{W\mathbf{v}^T}_{\text{Velocity of frame S}}, \underbrace{S\mathbf{t}_{C_1}^T, \mathbf{q}_{SC_1}^T}_{\text{Sensor extrinsics (here: camera)}}, \underbrace{S\mathbf{t}_{C_2}^T, \mathbf{q}_{SC_2}^T}_{\text{Sensor extrinsics (here: camera)}}, \underbrace{\mathbf{b}_g^T, \mathbf{b}_a^T}_{\text{Sensor intrinsics (here: IMU)}} \right]$$

Position and orientation of
frame S origin w.r.t. in W

Velocity of frame S
origin expressed in W

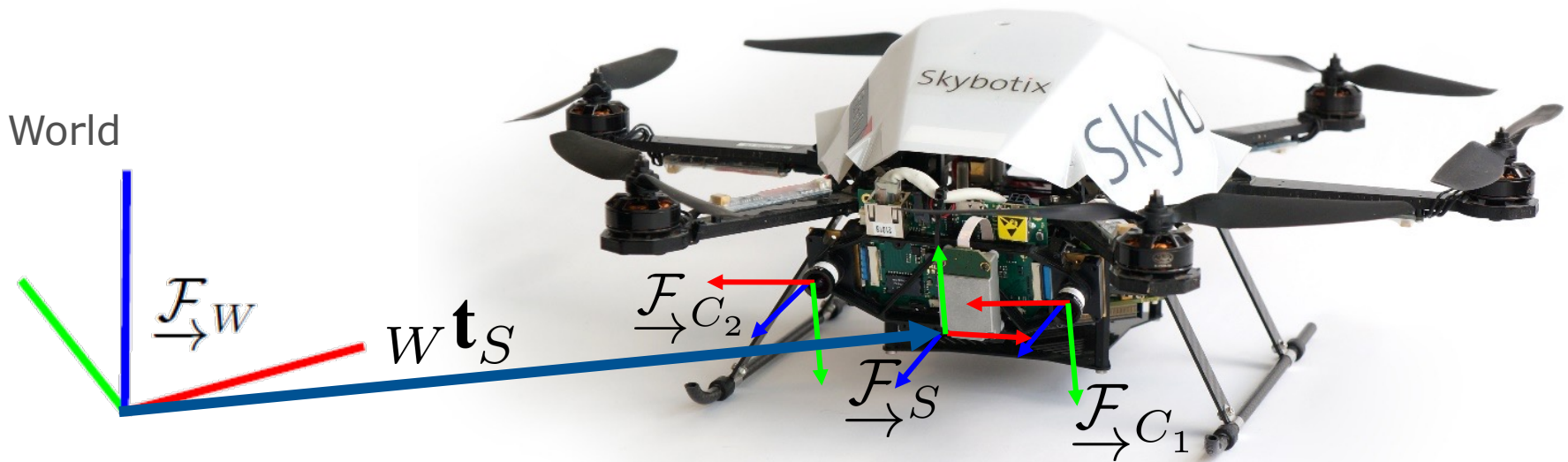
Poses of cameras 1 and 2
relative to frame S

IMU gyro and accelerometer
biases

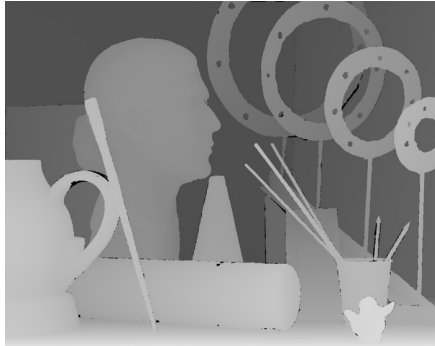
Physical robot states

**Sensor extrinsics
(here: camera)**

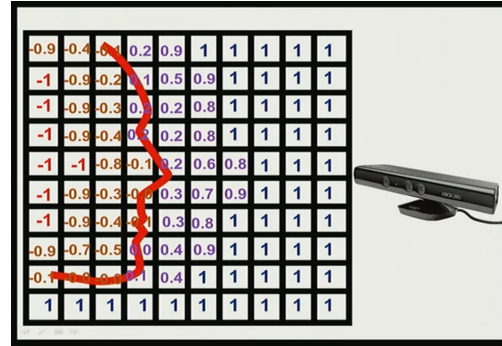
**Sensor intrinsics
(here: IMU)**



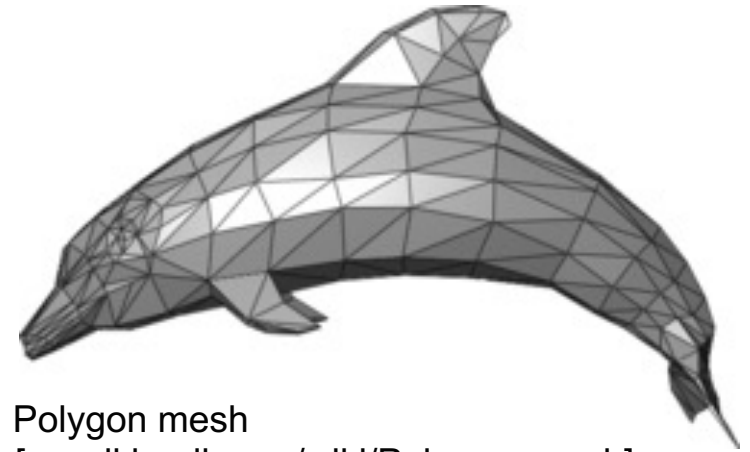
Map Representations



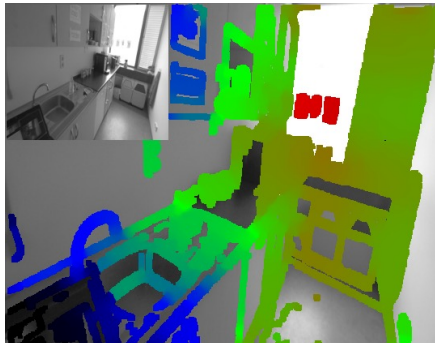
Depth maps
[vision.middlebury.edu]



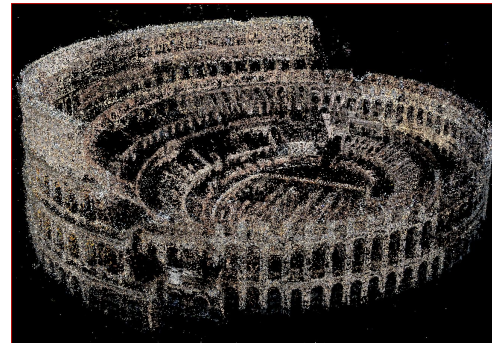
Truncated Signed Distance
Function [pointclouds.org]



Polygon mesh
[en.wikipedia.org/wiki/Polygon_mesh]



Semi-dense depth maps
[vision.in.tum.de]



Point clouds (here: sparse)
[grail.cs.washington.edu]



Surfel maps
[wp.doc.ic.ac.uk/thefutureofslam]

Probabilistic Estimation

We use Bayesian Statistics:

Measurements $\mathbf{Z}$ are modeled as samples from a **distribution** $p(\mathbf{z}|\mathbf{x})$ given the variables $\mathbf{x}$ (here: robot states plus possibly the map).

- **Maximum Likelihood (ML) Estimation:**

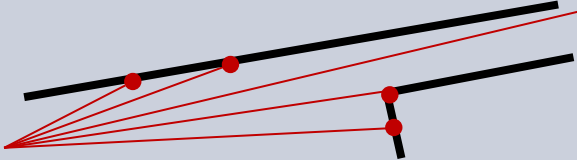
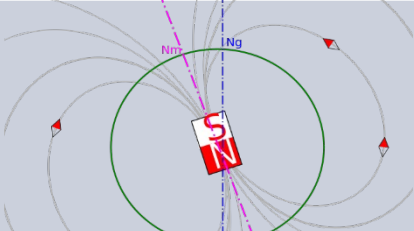
What values for variables best explain all the measurements?

- **Maximum a Posteriori (MAP) Estimation:**

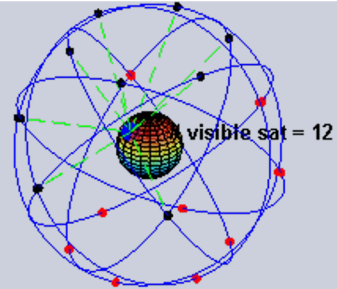
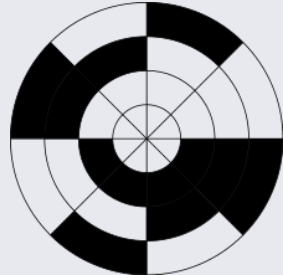
What values for variables best explain all the measurements and a prior belief $p(\mathbf{x})$ about those variables?

How to compute these values? To be discussed in later lectures.

Typical Sensors – Exteroceptive

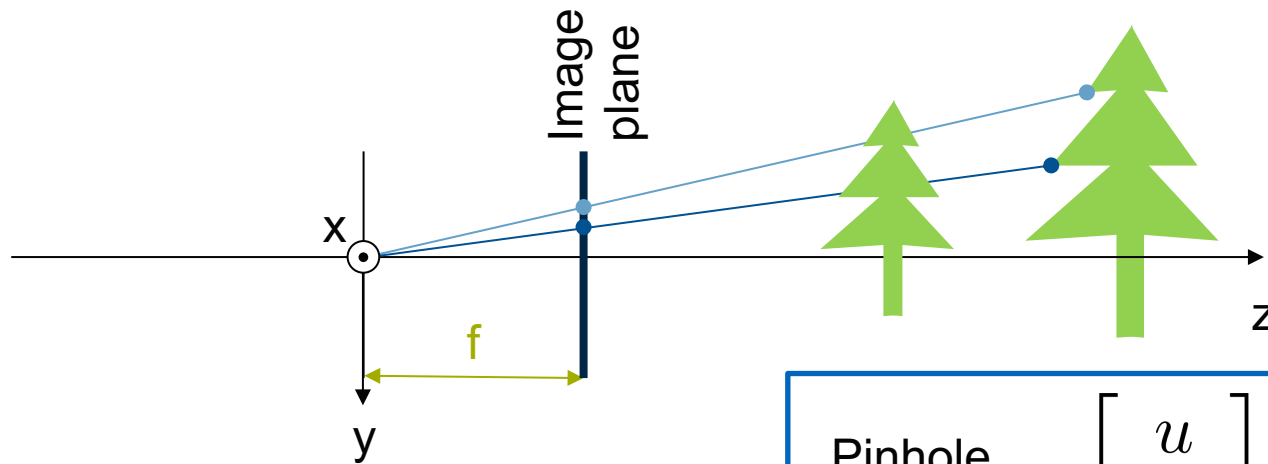
Sensor	Measurement	
Laser Scanner	3D points	
Camera	(Colour) image (RGB-D: with depth!)	
Magnetometer	3D magnetic field	
Pressure sensor	Air pressure (altitude / airspeed)	

Typical Sensors – Proprioceptive

Sensor	Measurement	
GPS	pseudo-ranges (position)	
Encoders	Joint / wheel angles	
Inertial Measurement Unit (IMU)	Rotation rates and accelerations (with caution: orientation)	

The Camera

**Simplified
pinhole model**



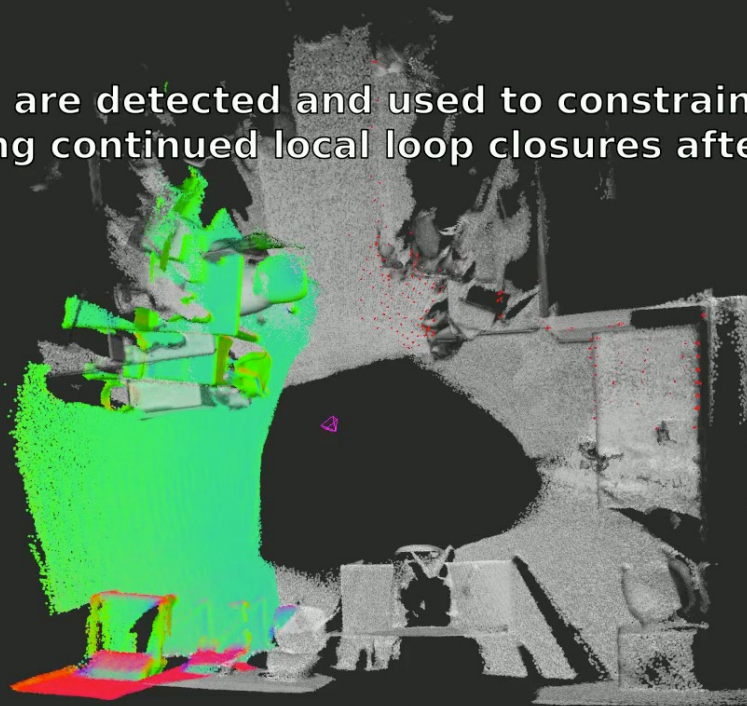
Pinhole projection:
$$\begin{bmatrix} u \\ v \end{bmatrix} = \frac{f}{z} \begin{bmatrix} x \\ y \end{bmatrix}$$

ElasticFusion [RSS'15, IJRR16]

Whelan, Salas-Moreno, Glocker, Davison, Leutenegger

Overview (Real-time)

Global loop closures are detected and used to constrain the surface globally enabling continued local loop closures afterwards

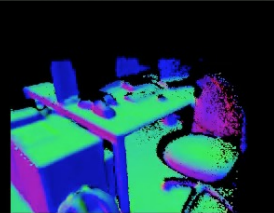
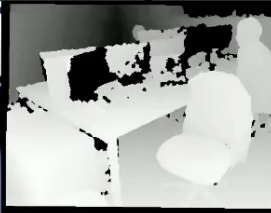


Live Colour

Live Depth

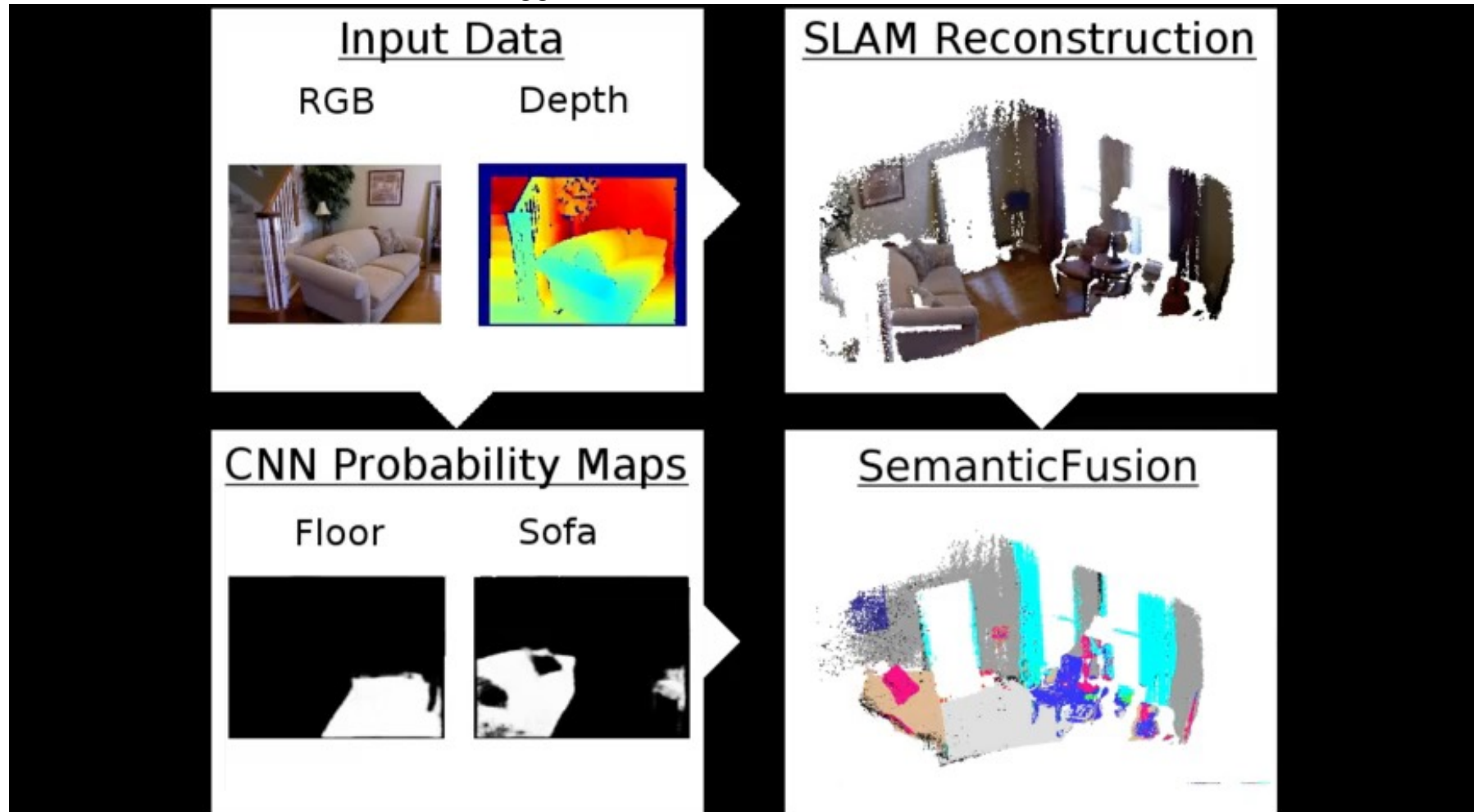
Active Colour

Active Surface



Fusing Semantics From CNN [ICRA'17]

McCormac, Handa, Davison, Leutenegger



Synthetic Data: Decrease Reality Gap [BMVC'18]

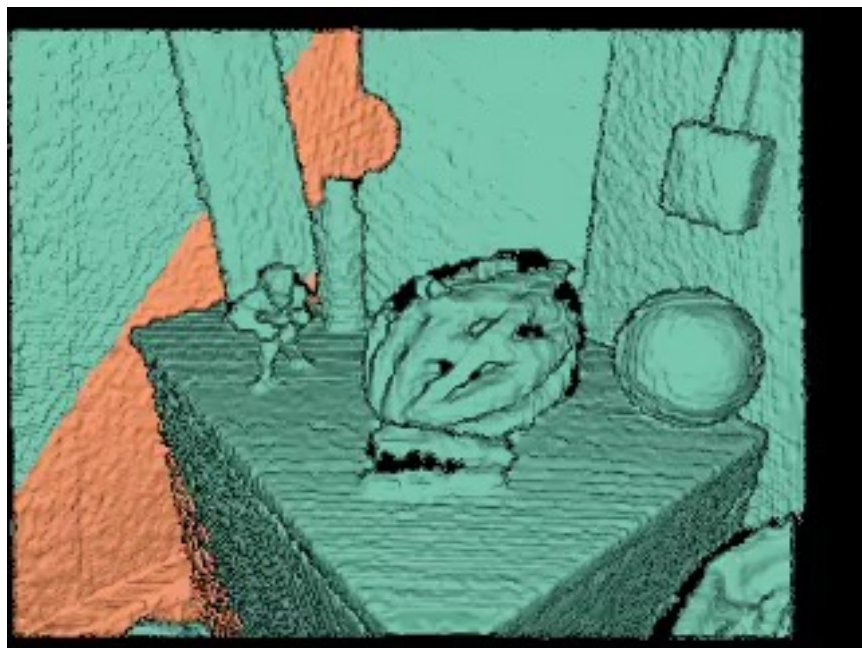
Li, McCormac, Clark, et al.



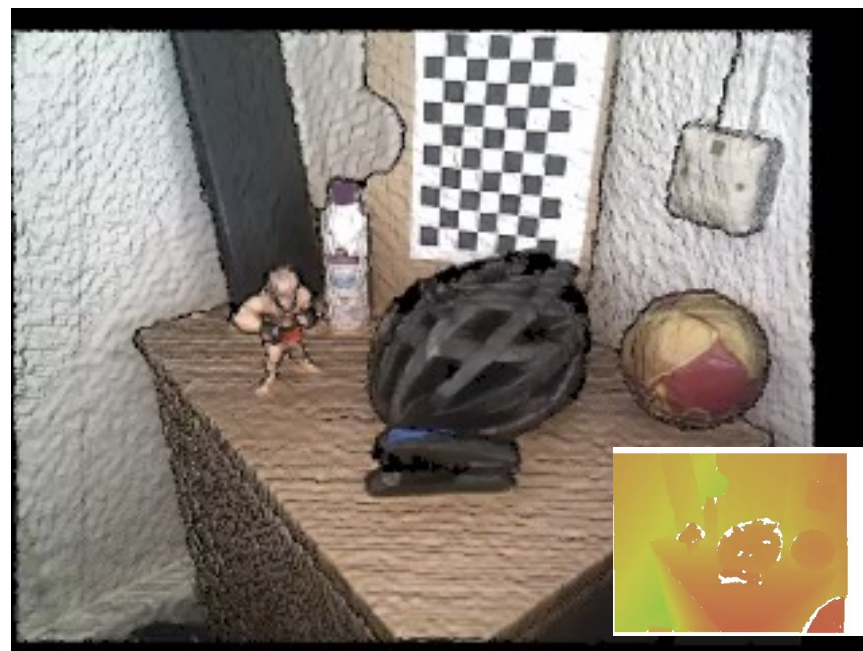
Live Reconstruction [3DV'19]

Vespa, Funk, Kelly, Leutenegger

3D reconstruction



3D RGB reconstruction



Raw depth

MID-Fusion [ICRA'19]

Xu, Tzoumanikas, Bloesch, Davison, Leutenegger



Color input

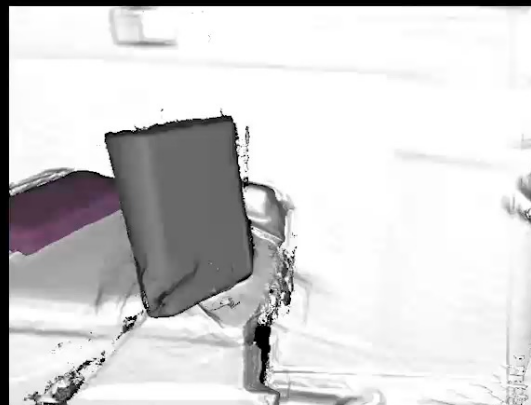


X1.5 speed

3D reconstruction



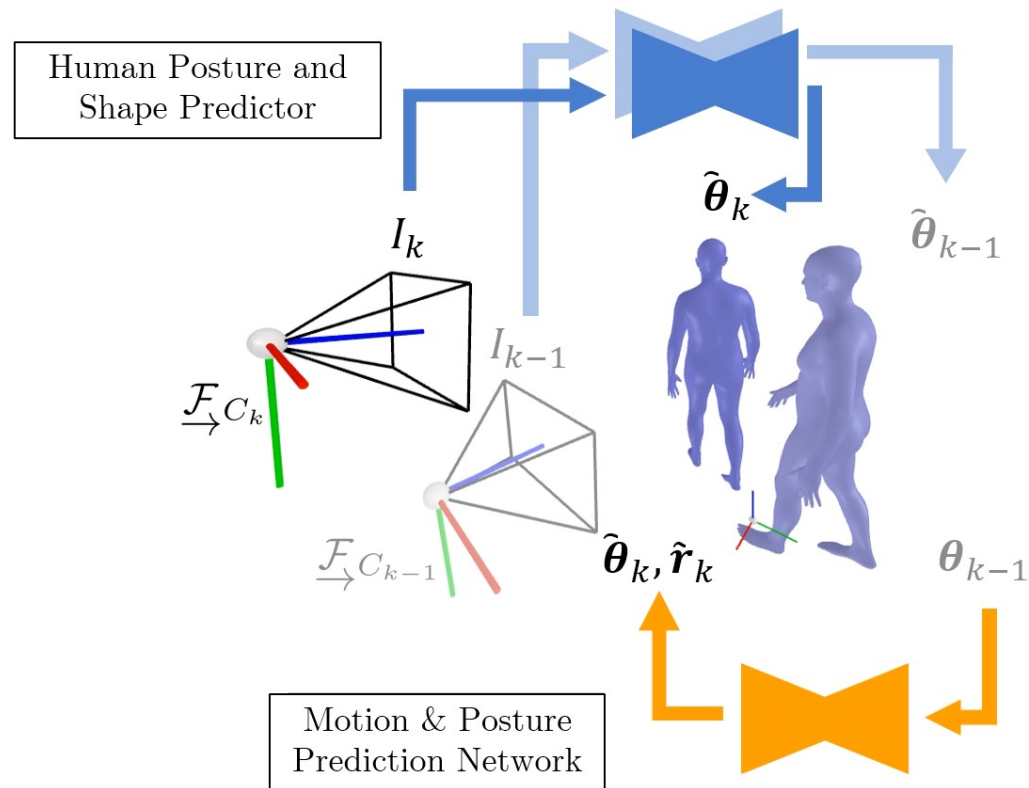
Depth input



3D label map

BodySLAM [ECCV'22]

Dorian Henning, Tristan Laidlow, Stefan Leutenegger

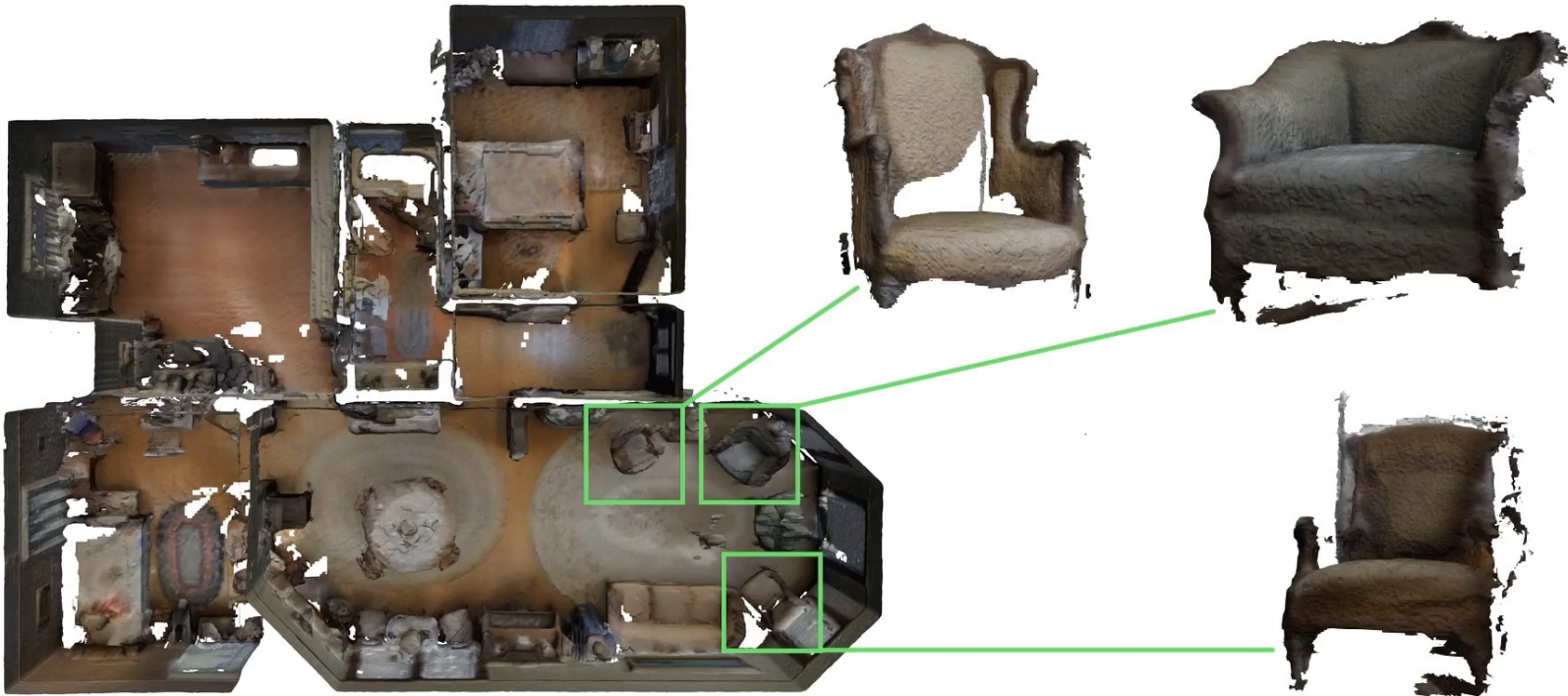


We introduce a human motion model, that uses past human postures and shapes to predict the next postures and scaled human centre pose changes.

Drone Semantic Exploration [ICRA'23]

Papatheodorou, Funk, Tzoumanikas, Choi, Xu, Leutenegger

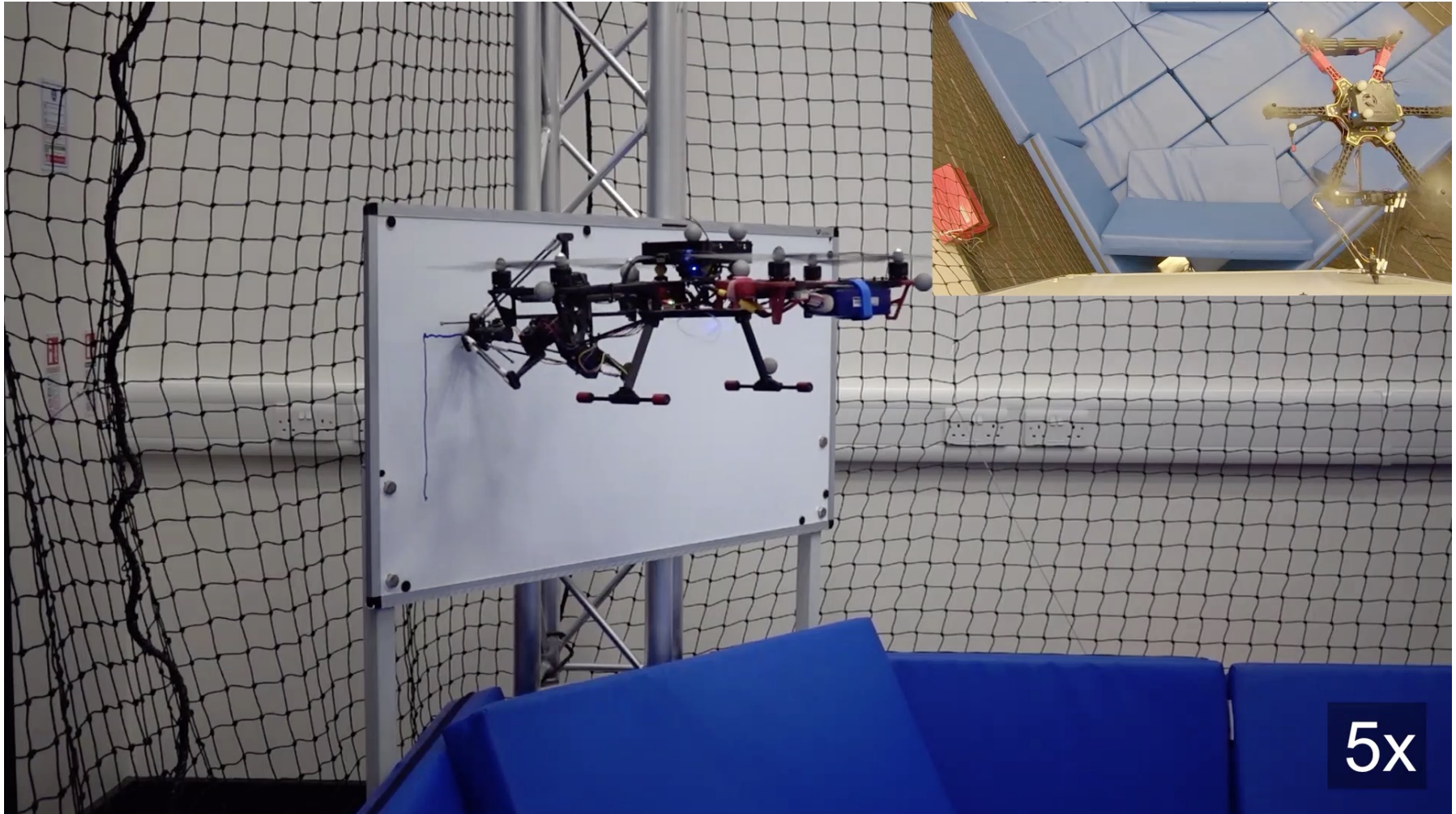
Object-centric autonomous exploration



We extend it to detect objects of interest and create detailed reconstructions

Towards Mobile Manipulation [RSS'20]

Tzoumanikas, Graule, Yan, Shah, Popovic, Leutenegger



Any Questions?

See you on **Friday at 12:00 for the exercises in 00.5901.051 (EE building)!**

... Or **tomorrow Thursday 10:00 for Mobile Robotics Practicals in 02.09.023**, for those who want!